

receives and thus the actual cost of providing care. This conversion is completed using hospital-specific cost-to-charge (CCR) ratios provided with NIS. Because CCRs are not available for NEDS, we apply average CCRs for each endpoint in NIS to the same set of ICD-10 codes in NEDS.

For each health endpoint, mean estimates are calculated using estimation commands for survey data to account for the sampling design and sample discharge weights of the HCUP data. This results in estimates of mean costs and a 95% confidence interval, which represents uncertainty in our valuation estimates of medical costs. The resulting estimates are presented in Table H-6.

Table H-6. Core Unit Values Available for Emergency Department Visits

Endpoint	ICD Codes	Age Range		Mean Unit Value (2015 \$)
		Min	Max	
ER visits, All Cardiac Outcomes	390-459	0	99	\$1,161
ER visits, respiratory	491-493, 460-466, 477, 480-486, 496, 786.07, 786.09	0	99	\$875

H.2.3 Emergency Room Visits for Asthma

Two unit values are currently available for use in BenMAP for asthma emergency room (ER) visits. One is \$533.69, from Smith et al., 1997, who reported that there were approximately 1.2 million asthma-related ER visits made in 1987, at a total cost of \$186.5 million, in 1987\$. The average cost per visit was therefore \$155 in 1987\$, or \$533.69 in 2015\$ (using the CPI for medical care to adjust to 2015\$). The uncertainty surrounding this estimate, based on the uncertainty surrounding the number of ER visits and the total cost of all visits reported by Smith et al. is characterized by a triangular distribution centered at \$533.69, on the interval [\$395.14, \$738.19].

A second unit value is \$446.52 from Stanford et al. (1999). This study considered asthmatics in 1996-1997, in comparison to the Smith et al. (1997) study, which used 1987 National Medical Expenditure Survey (NMES) data. In comparing their study, the authors note that the 1987 NMES, used by Smith et al., “may not reflect changes in treatment patterns during the 1990s.” In addition, its costs are the costs to the hospital (or ER) for treating asthma rather than charges or payments by the patient and/or third party payer. Costs to the ER are probably a better measure of the value of the medical resources used up on an asthma ER visit (see above for a discussion of costs versus charges).

The unit values and the corresponding distributions available in BenMAP for asthma-related ER visits are summarized in Table H-7.

Table H-7. Core Unit Values Available for Asthma-Related ER Visits

Basis for Estimate	Age Range		Unit Value (2015\$)	Distribution of Unit Value	Parameters of Distribution	
	Min	Max			P1	P2
COI: Smith et al. (1997)	0	99	\$534	Triangular	\$395	\$738
COI: Standford et al. (1999)	0	99	\$447	Normal	8.95	--

H.2.4 Emergency Hospital Admissions for All Respiratory

For HIFs based on studies that specifically assess the impact of pollutants on emergency hospital admissions (EHAs), we develop primary COI estimates based on a combination of hospitalization and emergency department visit costs using data from the Healthcare Cost and Utilization Project (HCUP). Within HCUP, we rely on both the 2016 Nationwide Emergency Department Sample (NEDS), and the 2016 Agency for Healthcare Research and Quality's Healthcare Utilization Project National Inpatient Sample (NIS) database for cost data. NEDS provides recent, nationally representative information on medical treatment in emergency departments, while NIS provides national hospital inpatient care data including charge information on all patients. Valuation estimates are based on the sum of total medical costs incurred in the emergency department and during the subsequent inpatient hospital stay plus the value of the lost productivity for time in hospital.

The NEDS and NIS datasets index cost data by medical billing codes. The principal diagnoses were used to define the health effects, see Section H.2.2 for additional details on health effect definition.

In the NIS dataset, we convert total charges (i.e., the amount billed to patients, employers, or insurance providers) into estimates of total costs (i.e., the final reimbursements for medical treatment) using hospital-specific cost-to-charge (CCR) ratios provided with NIS. Unadjusted charges are not suitable for use in regulatory analysis because posted prices generally do not reflect actual medical costs due, in part, to negotiation between medical providers and payers (e.g., insurance companies). We assume that adjusted charges reflect the actual revenue the hospital receives and thus the actual cost of providing care. Because CCRs are not available for NEDS, we apply average CCRs for each health effect in NIS to the same set of ICD-10 codes in NEDS.

For EHAs, mean cost estimates are calculated for both hospitalizations originating in the ER, via NIS data, and emergency department visits resulting in inpatient stays, via NEDS data, using estimation commands for survey data to account for the sampling design and sample discharge weights of the HCUP data. This results in estimates of mean costs and average length of hospital stay along with 95% confidence intervals, which represents uncertainty in our valuation estimates of medical costs and lost wages. The total COI for an emergency hospital stay lasting n days is estimated as the mean total ED and hospital charge plus n times the daily lost wage. Opportunity cost of a day spent in the hospital is estimated as the value of the lost daily wage, regardless of

whether or not the individual is in the workforce. Lost daily wages are described in more detail in Section H.2.1.

The resulting EHA estimates for all respiratory illnesses are presented in Table H-8. Although unit values available for use in BenMAP are county-specific, the national median daily wage was used to calculate opportunity costs and total costs.

Table H-8. Core Unit Values Available for Respiratory-related EHAs

Endpoint	ICD Codes	Age Range		Mean Hospital Charge (2015\$)	Mean Length of Stay (days)	Total Cost of Illness (2015\$)
		Min	Max			
EHA, All Respiratory	460-519	65	99	\$11,111	5.09	\$11,990

H.3 Other Health Endpoint Occurrence

Monetary valuation estimates for health endpoint occurrences other than hospital admissions or emergency department visits are described below, listed in alphabetical order.

H.3.1 Lung Cancer

The unit value for non-fatal lung cancer incidence is derived from the direct medical costs of lung cancer treatment estimated by Kaye et al. (2018). Lost earnings are assumed to be negligible because of the low labor force participation rate among the age groups at highest risk of developing lung cancer (average age of diagnosis is approximately 70 years). Lung cancer treatment costs depend to a large extent on the phase of care, with costs in the initial year of treatment (\$17,422 for males) far exceeding the continuing costs of treatment in subsequent years (\$3,269 for males). We calculated costs over a five-year span, beginning with the initial onset which occurs with a delay after exposure. The initial year's treatment cost is summed with four years of continuing annual costs discounted by 2%, 3% and 7%.

Furthermore, Kaye et al. (2018) provides separate treatment cost estimates for men and women. The distribution of new lung cancer cases by sex in the United States from Siegel et al. (2019) is approximately 51% male and 49% female. This distribution of new lung cancer cases was used to weight the sex-specific cost estimates from Kaye et al. (2018) to obtain a combined five-year cost estimate for both sexes. In order to adjust the cost estimate to 2015\$ using a medical cost index, we assume that costs presented by Kaye et al. (2018) are in 2010\$ as an approximate midpoint of the data years 2007-2012. Altogether, the cost of non-fatal lung cancer incidence over a five-year period is estimated to be \$34,155 using a 2% discount rate, \$33,809 using a 3% discount rate or \$32,548 using a 7% discount rate (Table H-9).

Table H-9. Core Unit Values Available for non-fatal Lung Cancer

Health Endpoint	Basis for Estimate	Age Range		Unit Value (2015\$)
		Min	Max	
Lung Cancer	COI: 5 yrs med, 2% DR, Kaye (2018)	30	99	\$34,155
	COI: 5 yrs med, 3% DR, Kaye (2018)	30	99	\$33,809
	COI: 5 yrs med, 7% DR, Kaye (2018)	30	99	\$32,548

For an outcome such as lung cancer, there is an expected time lag between changes in pollutant exposure in a given year and the total realization of health effect benefits, commonly referred to in regulatory analyses as the “cessation lag.” The time between exposure and diagnosis can be quite long, on the order of years to decades, to realize the full benefits of the air quality improvements. This latency period is important in order to properly discount the economic value of these health benefits.

To estimate the latency period, we performed a literature search using the keywords “non-fatal lung cancer,” “lung cancer,” “PM_{2.5},” “latency,” and “incidence.” Five papers that estimate the risk of lung cancer incidence from PM_{2.5} exposure using a latency period were identified. The latency period length and country of the identified papers are summarized in Table H-10. Based on estimates of lung cancer latency from the literature, 10 years was the most common latency period estimate found in the literature (i.e., the mode).

Table H-10. Latency Periods Used in Lung Cancer Risk Assessment Papers

Study	Latency Period (years)	Location
Gogna et al., 2019	5	Canada
Bai et al., 2020	4; 10	Canada
Kulhanova et al., 2018	10	France
Coleman et al., 2020	10; 15	US
Harrison et al., 2004	20	US

To account for the latency period between air pollution reductions and avoided lung cancer diagnoses in our economic valuation estimates, we developed an age-at-diagnosis cessation lag distribution method based on an approach previously used to estimate avoided cases of kidney cancer in analyses of water quality rules (U.S. EPA, 2017). The method uses lung and bronchus cancer diagnosis age-distribution from the Surveillance, Epidemiology, and End Results Program (SEER). For this model, we assumed that the case reduction distribution would follow the age-pattern of cancer diagnosis between the age at which the exposure change occurs and 99 years. Table H-11 shows an example case reduction distribution calculation for an exposure change experienced at 55. SEER estimates 92.2% of lung and bronchus cancer cases occur in individuals 55 years and older. Dividing the percentages in the remaining age bins by 92.2% (the percent of lung and bronchus cancer diagnoses between the age of exposure